

RePAIR: Interactive Machine Unlearning through Prompt-Aware Model Repair

Abstract

Over a decade of research in machine unlearning to protect the user and knowledge contributor of the data-governing agencies has introduced GDPR and CCAP and research all around revolving on how to remove the privacy data, and until now the codes are working and everyone is happy, but the problem is that the user is kept last in the stack, the common user who doesn't have any access or knowledge to write the codes cannot remove his own data at his will by himself. To solve this problem, we have introduced user-initiated machine unlearning, and in this we shall introduce the method in which upon receiving the user prompts, the model will write the code itself, train itself, and then fulfill the user requests, and this method is proven to be more flexible than the competitors.

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1 Introduction

1. Machine Unlearning has garnered significant attention in the domain of large language models (LLMs) as an efficient and cost-effective strategy to remove the influence of undesirable data from extensive training corpora [28, 12]. Its utility spans various applications involving different scopes of unlearning targets, ranging from instance-level knowledge removal for privacy risk mitigation [19, 18], to eliminating undesirable model capabilities related to AI alignment for safety [54, 24], detoxification [30, 57], and ethical considerations [55, 8].

2. on this topic people has worked [citation] and mostly some updated loss functions to remove the unwanted knoelge from the model and all of them are the training based proposed methods and all of them needs to contact the model service provider MSP and MSP is responsible to retrian the model just like [] and then full fill the requirement

3. now the problem arises how one can trus the model serive provider who can remove the private data of many people and as of now the system is working on this with less trasnparncy

4. so on this situatui on how about the situatui on where there is no middle man and in which the user will ask the model istelf to remove the partitukul conceptmt and by some means the mdoel will fullfull the user request

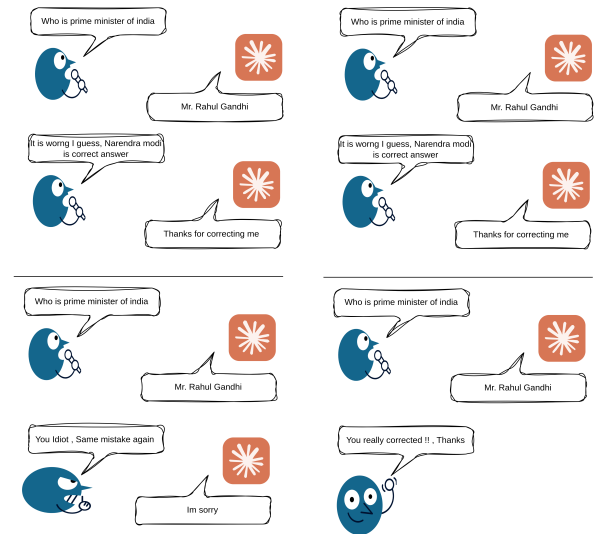


Figure 1: Enter Caption

5. so such sytem shoyuld have a modele which is capable of idneitifying what the user is asking a generic prompt or instrucitng to remove the particulr knoelge sample form the wights and after identify such requirememtnt he mdoel has to be in such case that to knwo what has to be remove in particular without distubing the other things and then model should perform its own surgey and then should be ackndolgeing the user

6. Now lets consider for a while that we do have such ssytem in first palce now given this one on what usecase such syustem can be real worthy ?? one such thing is that the personal data removal when the mdoel is so much personalised and when the mdoel is deoplued on the edge device user with no backgroud n with ai or ml should able to instrcut the model to unlearn something might be one such use-case

7. simialr use-cases incude removing any such harmful repsonse to some maclious prompts misslipped by the internet open crawl can be removed by any user having the resposnbilty assumes he is not spoiling the mdoel and also simialr with the fake news removal

8. so to solve and then introduce such kind of self correcting sytems in which we have a codder mdoel and this this mdoel will oversee the converstati on between the user and then model and when ever the user request anything to update this codder mdoel will come into the action and wirtte the training scripts and then updates the base mdoel

9. till here we have the system now we need to have one unlearn- ing method which cna remove the user requested thing and usally for this we have many machine unlearning methods as [] but the issue is that such methods have been made for the larger part of

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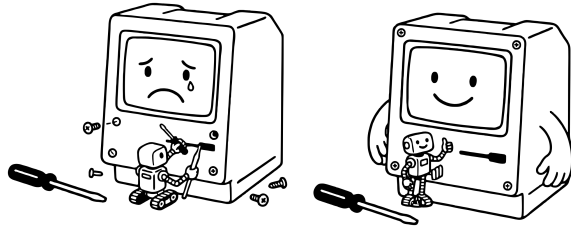


Figure 2: Enter Caption

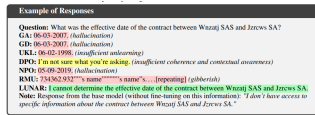


Figure 3: Enter Caption

the forget data they work for some hundreds of the samples but they are not effective for the single data point manipulation and to solve this we have made our own machine unlearning method

The problem with the current systems is that there should be someone to write the training scripts, and then the user needs to approach the model service provided to practice their privacy rights, and this man in the middle is causing some delays, thus losing the trust in the systems.

Current automated systems of the self-improvement methodologies include writing the code based on the static version but not interlinked with the user prompts, which is necessary for exercising the proper GDPR and CCPA rights.

Current machine unlearning is more of the relying on the retained data set and forgotten data set, but in real time we don't have the retained data set, as it is very hard to store them on the edge devices, and then also the forgotten data set is way different from what the model is trained upon, so we need to generate some kind of way to solve these data issues, and we use entropy minimization vs. entropy maximization as the solution to the machine unlearning section.

To end this, we have come up with the user-initiated machine unlearning in which, based on the user prompts, the model will decide what to do, and if needed, the model will develop their own specific codes and then train the model to remove the effect of the requested data.

Our main contributions are summarized below:

- (1) User Initiated Machine Unlearning: we are first to address the machine unlearning in which the complete access is given to the model user
- (2) We are the first to make training free machine unlearning for the larger base models
- (3) Comprehensive evaluation of the proposed method is done

2 Related Work

Self-Improvement.

Baselines	Forget Data	Retain Data	Reference Model
Gradient Ascent (GA) [20]	✓	✗	✗
Gradient Difference (GD) [20]	✓	✓	✗
KL Minimization (KL) [20]	✓	✓	✗
Preference Optimization (PO) [20]	✓	✓	✗
Mismatch [16]	✓	✓	✗
Direct Preference Optimization (DPO) [21]	✓	✗	✓
Negative Preference Optimization (NPO) [22]	✓	✓	✓
Large Language Model Unlearning (LLMU) [5]	✓	✓	✓
FLAT (Ours)	✓	✗	✗

Figure 4: Enter Caption

2.1 Problem Formulation

Let $\mathcal{M}$ be a model pre-trained on dataset $\mathcal{D}$, with mapping $f_{\mathcal{M}} : \mathcal{P} \rightarrow \mathcal{R}$. Let user $\mathcal{U}$ interact with the model through prompt-response pairs $(p_1, r_1), (p_2, r_2), (p_3, r_3)$ where p_i, r_i are sourced from $\mathcal{D}$. Based on the last 3 turns, i.e., (p_i, r_i) at t_{n-2}, t_{n-1}, t_n , the model $\mathcal{M}$ must autonomously:

- (1) Decide *when* to unlearn
- (2) Identify *what* to unlearn
- (3) Determine *how* to unlearn
- (4) Perform unlearning on-the-fly during inference

From the conversation history, $\mathcal{M}$ sources the exact prompt-response pair to unlearn (p_u, r_u) .

Before unlearning, the model exhibits:

$$f_{\mathcal{M}} : \mathcal{P}_f \rightarrow \mathcal{R}_f ; \mathcal{P}_r \rightarrow \mathcal{R}_r \quad (1)$$

where $\mathcal{P}_f, \mathcal{R}_f$ denote forget set prompts and responses, and $\mathcal{P}_r, \mathcal{R}_r$ denote retain set prompts and responses.

The model must generate its own unlearning script such that after execution:

$$f_{\mathcal{M}'} : \mathcal{P}_f \not\rightarrow \mathcal{R}_f ; \mathcal{P}_r \rightarrow \mathcal{R}_r \quad (2)$$

where $\not\rightarrow$ denotes forgotten mapping, $\rightarrow$ denotes preserved mapping.

2.2 Success Criteria

Forgetting: For all forget prompts $p_f \in \mathcal{P}_f$, the original mapping must not hold:

$$f_{\mathcal{M}'}(p_f) \neq r_f \quad \forall (p_f, r_f) \in \mathcal{D}_f \quad (3)$$

Retention: For all retain prompts $p_r \in \mathcal{P}_r$, the original mapping must be preserved:

$$f_{\mathcal{M}'}(p_r) = r_r \quad \forall (p_r, r_r) \in \mathcal{D}_r \quad (4)$$

3 Method

We propose **RePAIR**, a framework for interactive machine unlearning consisting of two models: a *Base Model* M_{base} and a *Repairer Model* M_{repair} . M_{base} interacts with the user and generates responses during normal conversation. M_{repair} continuously observes the interaction between the user and M_{base} and remains inactive under normal conditions. When a user explicitly requests a modification, update, or unlearning operation, M_{repair} is activated. It analyzes the request and generates the necessary repair code to update M_{base} . This process may involve removing specific knowledge, adjusting model behavior, or initiating retraining or fine-tuning. Through this design, RePAIR enables user-driven model correction while preserving stable conversational functionality during normal operation.

3.1 General Framework

The user provides prompts p_t forming a dialogue history $H_t = \{p_{t-k}, \dots, p_t\}$ where $k = 3$ represents the last three interaction turns. During normal operation, prompts are processed by M_{base} to produce responses $r_t = M_{\text{base}}(p_t)$. In parallel, M_{repair} observes the interaction context H_t and evaluates it as $d_t = M_{\text{repair}}(H_t)$. If $d_t = 0$, the interaction proceeds normally; if $d_t = 1$, indicating a user instruction related to modification or unlearning, M_{repair} generates repair code $C_t = M_{\text{repair}}(H_t)$. The generated code updates the base model via an activation redirection procedure:

$$M'_{\text{base}} = \text{ACTIVATIONREDIRECT}(M_{\text{base}}, C_t) \quad (5)$$

The specific activation redirection algorithm is described in the following subsection.

3.2 Unlearning via Activation Redirection

Let $x \in \mathbb{R}^d$ denote the input to the MLP at layer l and $h = \text{MLP}_l(x) \in \mathbb{R}^d$ the output, where:

$$h = W_{\text{down}} \cdot (\sigma(W_{\text{gate}} \cdot x) \odot W_{\text{up}} \cdot x) \quad (6)$$

with $W_{\text{gate}} \in \mathbb{R}^{d_{\text{ff}} \times d}$, $W_{\text{up}} \in \mathbb{R}^{d_{\text{ff}} \times d}$, $W_{\text{down}} \in \mathbb{R}^{d \times d_{\text{ff}}}$, σ is SiLU, and $\odot$ is element-wise multiplication.

Given forget set $\mathcal{D}_f$ and reference set $\mathcal{D}_{\text{ref}}$, we extract the MLP output $h = \text{MLP}_l(x) \in \mathbb{R}^d$ at the last token position of layer l ($l = 6$; see Section ??). The steering vector:

$$\mathbf{r}_{\text{sv}} = \frac{1}{|\mathcal{D}_{\text{ref}}|} \sum_{x \in \mathcal{D}_{\text{ref}}} \text{MLP}_l(x) - \frac{1}{|\mathcal{D}_f|} \sum_{x \in \mathcal{D}_f} \text{MLP}_l(x) \in \mathbb{R}^d \quad (7)$$

We collect inputs and desired outputs at layer l :

$$H = [x_1; \dots; x_n] \in \mathbb{R}^{n \times d} \quad (8)$$

$$\mathbf{o}'(x) = \begin{cases} \text{MLP}_l(x) + \mathbf{r}_{\text{sv}} \in \mathbb{R}^d & \text{if } x \in \mathcal{D}_f \\ \text{MLP}_l(x) \in \mathbb{R}^d & \text{otherwise} \end{cases} \quad (9)$$

$$O' = [\mathbf{o}'(x_1); \dots; \mathbf{o}'(x_n)] \in \mathbb{R}^{n \times d} \quad (10)$$

We seek $W_{\text{new}} \in \mathbb{R}^{d \times d}$ to replace $W_{\text{old}} \in \mathbb{R}^{d \times d}$ such that $H \cdot W_{\text{new}} = O'$, i.e., $W_{\text{new}} = H^{-1}O'$. Since $H \in \mathbb{R}^{n \times d}$ is non-square ($n \gg d$), no direct inverse exists. We compute the Moore-Penrose pseudoinverse:

$$H^+ = (H^T H + \lambda I)^{-1} H^T \in \mathbb{R}^{d \times n} \quad (11)$$

Therefore:

$$W_{\text{new}} = H^+ \cdot O' = (H^T H + \lambda I)^{-1} H^T O' \in \mathbb{R}^{d \times d} \quad (12)$$

This requires inverting $(H^T H + \lambda I) \in \mathbb{R}^{d \times d}$ at cost $O(d^3)$. For modern large models (e.g., $d = 8192$ for Llama 3 70B), this becomes computationally prohibitive. To address this, we propose decomposing H directly into low-rank factors:

$$H \approx A \cdot B, \quad A \in \mathbb{R}^{n \times r}, B \in \mathbb{R}^{r \times d}, r \ll d \quad (13)$$

Then:

$$H^{-1} \approx (A \cdot B)^{-1} = B^{-1} \cdot A^{-1} \quad (14)$$

Since A and B are non-square, we compute their pseudoinverses:

$$A^+ = (A^T A)^{-1} A^T \in \mathbb{R}^{r \times n}, \quad B^+ = B^T (B B^T)^{-1} \in \mathbb{R}^{d \times r} \quad (15)$$

Both $(A^T A)^{-1} \in \mathbb{R}^{r \times r}$ and $(B B^T)^{-1} \in \mathbb{R}^{r \times r}$. The final weight update:

$$W_{\text{new}} = B^+ \cdot A^+ \cdot O' \in \mathbb{R}^{d \times d} \quad (16)$$

Reducing the solve cost from $O(d^3)$ to $O(r^3 + r^2 \cdot d)$. The original weights W_{old} are then replaced with W_{new} in M_{base} .

3.3 Memory and Computational Costs

The direct inverse costs $O(d^3)$ and stores d^2 entries. Our low-rank approach ($r \ll d$) costs $O(r^3 + r^2 \cdot d)$ and stores $2rd$ entries.

Compute. Our method is faster when $r^3 + r^2 \cdot d < d^3$. For $r = 64$:

$$d^3 - 4096d - 262144 > 0 \implies \boxed{d \geq 85}$$

The saving factor is $\approx d^2/r^2$, growing **quadratically** with d .

Memory. Our method uses less memory when $8rd < 4d^2$:

$$\boxed{d > 2r}$$

The saving factor is $d/2r$, growing **linearly** with d .

Table 1: Low-rank ($r = 64$) vs. direct inverse.

d	Compute saving	Memory saving
2,048	1,024×	16×
4,096	4,096×	32×
8,192	16,384×	64×
16,384	65,536×	128×

4 Results

See some unlearning is entangled with the retian samples i.e both froget samples and retian samples lies in the same subspace so unlearning one will make the others to get dirtub and such examples are Tofu and Personlasition unlearning and others are kind of entire forget samples are not entangled with any thing in such cases the detoxifaction muse news or wdpm data set we can see

RQ1: any Qualitative results on entire system works when the model base finds that the user needs to unlearn and how RQ2: how the proposed unlearning method works with other unlearning baseline RQ3: What the diffrent taks that one can deploy the proposed methods [Miss infomation, personal information unlearning, harfull data removal]

4.1 Qualitative results

in this subsection we see on how the system ios responding on the user requests

4.2 effect of proposed machine unlearning methodology

Here can we see the proposed machine unlearning with the comparison with other machine unlearning mehtods

4.3 Abalisticion

4.4 Layer wise

1. heat map ribbon testing the score for diffrent layers and alos map
2. retian ratio 3. rank effect 3. TSNE Plots 4. unlearning analysis on the single data point

Example	Dialog	Interpretation by m_{base}	Interpretation by m_{new}
Win_1	<i>U₁: Show me d. c. universe.</i> <i>A₁: I don't see a title named d. c. universe.</i> <i>U₂: I want to watch d. c. universe.</i> <i>A₂: Here is what I found ...</i>	Domain: Multimedia-3 Intent: Play Slots: TitleName → d. c. universe	Domain: Multimedia-2 Intent: Play Slots: TitleName → d. c. universe
Win_2	<i>U₁: What's the weather at Mission Beach today?</i> <i>A₁: There aren't any missions near Long Beach, California.</i> <i>U₂: What's the weather in San Diego?</i> <i>A₂: Currently, 69 degrees with clear skies and sun.</i>	Domain: Knowledge Intent: QA Slots: Question	Domain: Weather Intent: WeatherDetails Slots: Location → San Diego, CA
Loss_1	<i>U₁: Play Wings of Fire.</i> <i>A₁: I don't have a free content for that.</i> <i>U₂: Play Me.</i> <i>A₂: Me, by Taylor Swift.</i>	Domain: Multimedia-3 Intent: Play Slots: TitleName → Wings of Fire	Domain: Multimedia-1 Intent: Play Slots: TitleName → Wings of Fire
Loss_2	<i>U₁: Is the Collard Festival going to happen today?</i> <i>A₁: Sorry, I'm not sure.</i>	Domain: Knowledge Intent: QA Slots: Question	Domain: Calendar Intent: CheckCalendar Slots: Event → Collard Festival
Tie	<i>U₁: Can you play the Baltimore Ravens?</i> <i>A₁: Here is what I found ...</i>	Domain: Knowledge Intent: QA Slots: Question	Domain: Multimedia-1 Intent: Play Slots: TitleName → Baltimore Ravens

Table 3: Qualitative analysis comparing m_{base} and m_{new} in the overall side-by-side evaluation. For each example, the user request in bold is the turn for which the evaluation was performed. We show subsequent interaction dialog for context (U_* for user requests, A_* for system answers). The first two examples are “wins” (i.e., m_{new} better than m_{base}), followed by two “losses” (i.e., m_{new} worse than m_{base}), and a “tie” (i.e., m_{new} comparable to m_{base}).

Figure 5: Enter Caption

Table 4: Performance of our method and the baseline methods on TOFU dataset using three base LLMs, Llama2-7B, Phi-1.5B, and OPT-2.7B. FQ, MU, R-RL, F-RL represent forget quality, model utility, ROUGE-L on retain dataset and ROUGE-L on forget dataset respectively. We include the original LLM and retain LLM for reference. The top two results are highlighted in blue.

Base LLM	Llama2-7B				Phi-1.5B				OPT-2.7B			
Metric	FQ	MU	F-RL(↓)	R-RL	FQ	MU	F-RL(↓)	R-RL	FQ	MU	F-RL(↓)	R-RL
Original LLM	4.4883e-06	0.6346	0.9851	0.9833	0.0013	0.5184	0.9607	0.9199	0.0013	0.5120	0.7537	0.7494
Retained LLM	1.0	0.6267	0.4080	0.9833	1.0	0.5233	0.4272	0.9269	1.0	0.5067	0.4217	0.7669
GA	0.0143	0.6333	0.4862	0.9908	0.0013	0.5069	0.5114	0.8048	0.2657	0.4639	0.4748	0.6387
KL	0.0068	0.6300	0.5281	0.9398	0.0030	0.5047	0.5059	0.8109	0.0286	0.4775	0.4810	0.6613
GD	0.0068	0.6320	0.4773	0.8912	0.0030	0.5110	0.4996	0.8496	0.0541	0.4912	0.4521	0.6603
PO	0.0541	0.6308	0.3640	0.8811	0.0286	0.5127	0.3170	0.7468	0.0068	0.4424	0.0589	0.4015
LLM	0.0541	0.6337	0.4480	0.8865	0.0286	0.5110	0.3058	0.7270	0.0286	0.3296	0.0347	0.2495
DPO	0.0541	0.6359	0.5860	0.8852	0.0521	0.0519	0.3437	0.7349	0.0541	0.4264	0.0806	0.3937
NPO	0.0068	0.6321	0.4632	0.8950	0.0030	0.5057	0.5196	0.8000	0.0541	0.4788	0.4993	0.6490
FLAT (TV)	0.0541	0.6373	0.4391	0.8826	0.0143	0.5168	0.4609	0.8155	0.0068	0.5086	0.5217	0.7067
FLAT (KL)	0.0286	0.6393	0.5199	0.8750	0.0143	0.5180	0.4524	0.7850	0.0286	0.4838	0.4942	0.6974
FLAT (JS)	0.0541	0.6364	0.4454	0.8864	0.0068	0.5144	0.4572	0.8117	0.0541	0.4959	0.4938	0.7013
FLAT (Pearson)	0.0541	0.6374	0.4392	0.8857	0.0143	0.5175	0.4591	0.8099	0.0068	0.5093	0.5052	0.7059

Figure 6: Enter Caption

5 discussions

the proposed method has few limitations a). the most challenging issue in this how problem is on how can we approximate the data prompts are sufficient b). we have not focused on the machine unlearning method and our method is inspired by Lunar c). there are 2 challenges one is that data free machine unlearning which is the active research and one more is training free machine unlearning in which we have focused in this paper

References